

Design and Development of In-Memory Computing Cell

Minor Project Report

*Submitted in Partial Fulfillment of the
Requirements for the Degree of*

MASTER OF TECHNOLOGY

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

(VLSI DESIGN)

By

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April 2025

CERTIFICATE

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Hardik Patel

Abstract

The increasing demand for faster and more energy-efficient computing has highlighted the inefficiencies in conventional systems, where the physical separation between processing and memory units leads to data transfer bottlenecks. This project addresses this challenge by implementing an In-Memory Computing (IMC) approach, which enables logic operations to be executed directly within memory arrays, thus minimizing unnecessary data movement.

The core of this study is the design and simulation of a novel memory cell that supports both data storage and logic operations such as NAND, NOR, XNOR, and implication. Built using CMOS technology, the proposed design is validated using industry-grade simulation tools to ensure it preserves traditional memory functionality while offering computational capabilities.

Performance metrics such as power usage, delay, and stability were assessed. The results show that embedding logic functions within memory structures can lead to notable improvements in energy efficiency and area usage, particularly in applications involving frequent or parallel operations.

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Chapter 1

INTRODUCTION

1.1.Introduction

Modern digital systems increasingly rely on efficient data handling, especially in scenarios demanding high throughput and low energy consumption. However, traditional architectures often struggle due to the overhead introduced by constant data transfers between separate memory and processor blocks. This bottleneck limits system performance and escalates power demands.

In-memory computing presents a compelling alternative by integrating data storage and processing within the same hardware unit. Leveraging SRAM structures, particularly the 8T variant, enables direct execution of logic operations inside memory without offloading computation to external units. This project explores such an implementation, aiming to streamline computing by integrating logic functions into the memory fabric itself.

1.2. Motivation

The persistent challenge in digital hardware is the delay and power consumption caused by shuttling data between memory and compute units. As system complexity grows, especially in domains like AI and real-time systems, these inefficiencies become more pronounced.

This project is motivated by the potential of in-memory computing to mitigate this problem. Specifically, 8T SRAM cells—thanks to their distinct read and write paths—offer a platform that supports in-place computation. The aim is to explore how fundamental Boolean and arithmetic functions can be realized within such a memory array, thus reducing external processing requirements and enhancing energy efficiency.

1.3. Problem Statement

Traditional von Neumann architectures separate processing and memory, leading to performance drawbacks from constant data exchange. This is particularly problematic in data-intensive applications, where it causes increased latency and power drain.

This work proposes an in-memory computing architecture using 8T SRAM cells to eliminate these inefficiencies. The system is designed to support Boolean logic and basic arithmetic directly within memory arrays. However, implementing computation within memory introduces new challenges, such as maintaining signal fidelity, optimizing sensing schemes, and ensuring low-power operation. This project addresses these issues through custom circuit design and simulation.

1.4. Objective

(1) Design a 8T SRAM cell

Ensure proper data retention and read/write functionality while targeting low-power applications.

(2) Developing sensing techniques for in-memory computing

Design efficient sensing mechanisms, including sense amplifiers and reference tuning, to detect bit-line discharge for logic interpretation.

(3) Implementing Boolean operations using 8T SRAM cells

Use the memory structure to carry out operations like AND, OR, XOR, etc., by exploiting bit-line behaviors.

(4) Simulate all design using Cadence Virtuoso

Model and test all circuits using Cadence Virtuoso to evaluate timing, voltage levels, and overall correctness.

(5) Analyze performance such as delay, power consumption

Evaluate power usage, delay, and signal integrity to benchmark the design against conventional methods.

1.5. Approach

The project follows a systematic design and verification process to implement and evaluate in-memory logic and arithmetic operations using 8T SRAM cells. The approach can be outlined in the following steps:

1) Selection of 8T SRAM Architecture

The 8-transistor SRAM cell was chosen due to its separate read and write paths, which provide improved read stability and enable concurrent access for multiple word lines. This characteristic is particularly advantageous for in-memory computing, as it allows the bit-line discharge to reflect logical combinations of stored values.

2) Logic Function Implementation within SRAM

Basic logic operations (AND, OR, NAND, NOR, XOR) were realized by enabling multiple word lines simultaneously and monitoring the resulting changes on the read bit-line (RBL). Custom sensing schemes and reference voltages were designed to detect the output logic level based on the voltage drop across RBLs.

3) Design Multi Logic Sense Amplifier(MLSA)

To ensure accurate and reliable detection of logic states, a voltage-reference-based sensing mechanism was designed. Multi Logic Sense Amplifiers were used to amplify small voltage differences on RBLs, with careful tuning of the reference voltage to separate different logic cases.

4) Schematic and Layout Simulation in Cadence

All designs were implemented using Cadence Virtuoso, and simulations were carried out using Spectre simulator at the 180nm CMOS technology node. This included schematic-level simulations of logic gates and arithmetic blocks under various input conditions, as well as transient analysis to observe timing and power behavior.

5) Perform Analysis.

Key performance metrics such as read/write stability, delay, energy consumption, and sensing margin were extracted from the simulation results. The performance of the implemented 8T SRAM-based IMC circuits.

1.6. Scope of the Project

This project explores the integration of logic operations within memory by combining 8T SRAM cells with a unified sensing circuit known as the Multi-Logic Sense Amplifier (MLSA). The aim is to perform multiple Boolean functions—including AND, OR, XOR, and their complements—within a single memory access cycle, using a simplified, low-power architecture. The 8T SRAM's decoupled read/write paths enable stable sensing of bit-line voltage variations, which are interpreted by the MLSA to determine logic outcomes. By adjusting reference voltages, the same hardware supports different logic operations without added complexity. The work involves schematic and layout-level design using Cadence tools at the 180nm technology node. Performance is validated through simulations focusing on power, delay, and functional accuracy. The project demonstrates a scalable approach for combining memory and computation in a compact design, offering potential for future energy-efficient processors.

Chapter 2

BACKGROUND & LITERATURE REVIEW

With the rising complexity of data-intensive applications, the limitations of traditional von Neumann architectures have become increasingly apparent, particularly due to the latency and energy costs associated with frequent data movement between separate memory and processing units. In-memory computing (IMC) has emerged as a promising alternative that mitigates this issue by allowing logical operations to be carried out directly within memory arrays, reducing data traffic and improving system efficiency.

Initial efforts in this domain utilized conventional 6T SRAM cells for in-memory logic operations such as AND, OR, and XOR. These implementations relied on activating multiple word lines and interpreting the resulting voltage changes on the shared bit-lines. However, 6T cells have limited read stability due to their shared read/write paths, which can lead to data corruption during simultaneous operations, making them less reliable for IMC.

To overcome these shortcomings, enhanced SRAM topologies like the 8T and 8+T cells were developed. These designs separate the read and write operations by introducing additional transistors, thereby improving stability and enabling more robust logic execution within memory. Researchers such as Dong et al. and Rajput et al. have demonstrated the feasibility of executing basic Boolean functions and arithmetic operations using 8T SRAM arrays in conjunction with voltage-based sensing techniques. These designs typically leverage the voltage drop on the read bit-line (RBL) and compare it against predefined reference levels to identify logic states.

Despite these advancements, many early IMC implementations relied on sense amplifiers customized for specific logic operations. While effective, this approach increases hardware complexity, area usage, and energy consumption. To address this inefficiency, a new approach called the Multi-Logic Sense Amplifier (MLSA) was introduced.

Chang and Chen proposed the MLSA as a unified sensing solution capable of performing multiple logic functions—such as OR/NOR, AND/NAND, and XOR/XNOR—using a single circuit. This design incorporates a voltage-tuned sensing mechanism and specialized transistor configurations to differentiate

between logic states based on subtle voltage variations. Their architecture also integrates a simple multiplexer for routing the desired output back into the memory array, enabling efficient write-back of computed results. Simulation results using 180nm CMOS technology revealed that the MLSA not only reduces sensing delay and power but also simplifies circuit integration.

Overall, these developments illustrate a significant shift toward memory-centric computing, where SRAM-based IMC architectures, especially those using multi-functional sensing units like the MLSA, offer scalable and energy-efficient solutions for future high-performance applications.

Chapter 3

METHODOLOGY

This section outlines the design approach used to develop an in-memory computing cell based on 8T SRAM architecture and a multi-functional sense amplifier. The methodology is broken down into key design stages, simulations, and performance analysis using industry-standard tools.

3.1 Design of 8T SRAM Cell Architecture

An 8-transistor SRAM cell was selected due to its improved stability and ability to support logic operations without disturbing stored data. This configuration separates the read and write paths, preventing unintended access during read cycles. The cell was constructed and verified using Cadence Virtuoso, targeting a 180nm CMOS process.

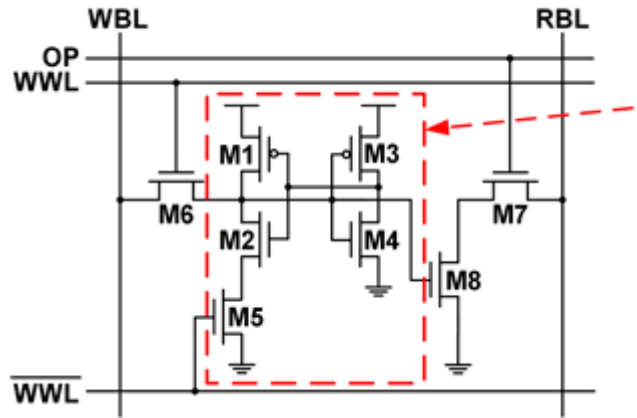


Figure 1: 8T SRAM Cell

3.2 Bitline-Based Logic Operation Setup

To implement logic functions, word lines corresponding to different memory cells were enabled at the same time. This caused varying discharge behaviors on the read bit-line depending on the input data stored in each cell. The voltage drop across the bit-line was monitored and used to infer logic outcomes.

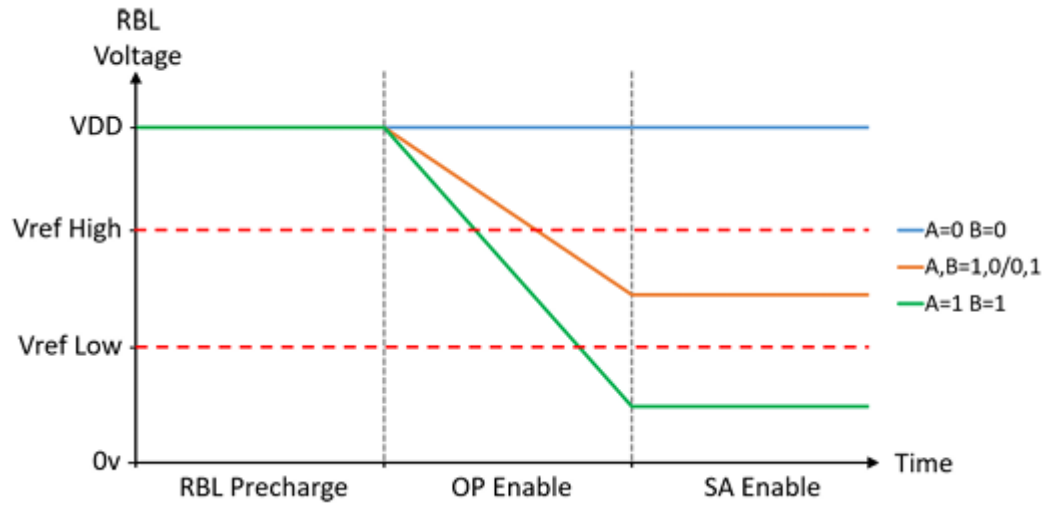


Figure 2: Bit-line discharge curve

3.3 Write-Back Operation

After a logic result is generated, it is passed to the write path using a controlled switching network. This enables the computed output to be stored back into the memory array in real time, supporting a full compute-in-memory cycle without external data handling.

3.4 Design of Multi Logic Sense Amplifier (MLSA)

A sense amplifier capable of identifying multiple logic conditions was designed. The amplifier uses a voltage-controlled latch circuit and additional logic to differentiate between logic states like OR, NOR, and XNOR. By tuning the reference voltage applied during sensing, the amplifier selectively recognizes the correct logic outcome.

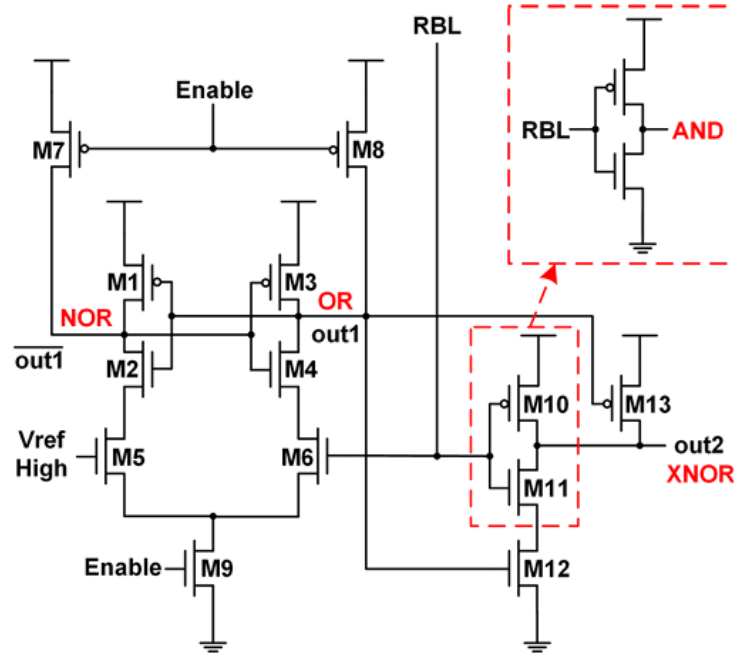


Figure 3: Multi Logic Sense Amplifier

Table 1: Interpreting Logic Outputs Using the Sense Amplifier

A	B	RBL Voltage	out1 (OR)	~out1 (NOR)	out2 (XNOR)
0	0	$V_{rbl} > V_{ref}$	0	1	1
0	1	$V_{DD}/2 < V_{rbl} < V_{ref}$	1	0	0
1	0	$V_{DD}/2 < V_{rbl} < V_{ref}$	1	0	0
1	1	$V_{rbl} < V_{DD}/2$	1	0	1

3.5 Simulation and Verification in Cadence

All components were simulated using the Spectre simulator in Cadence at 180nm. Transient simulations validated the behavior of each logic function under different input states. Key metrics such as voltage response, timing, and power consumption were extracted to evaluate design performance and robustness.

Chapter 4

HARWARE AND LAYOUT DESIGN

4.1 8T SRAM Cell Design

The foundation of the in-memory computing architecture is the 8-transistor SRAM cell. This design includes six transistors for the storage core and two additional transistors to create an isolated read port. This decoupling of read and write paths allows the simultaneous activation of multiple word lines without corrupting stored data, making it suitable for voltage-based logic computation

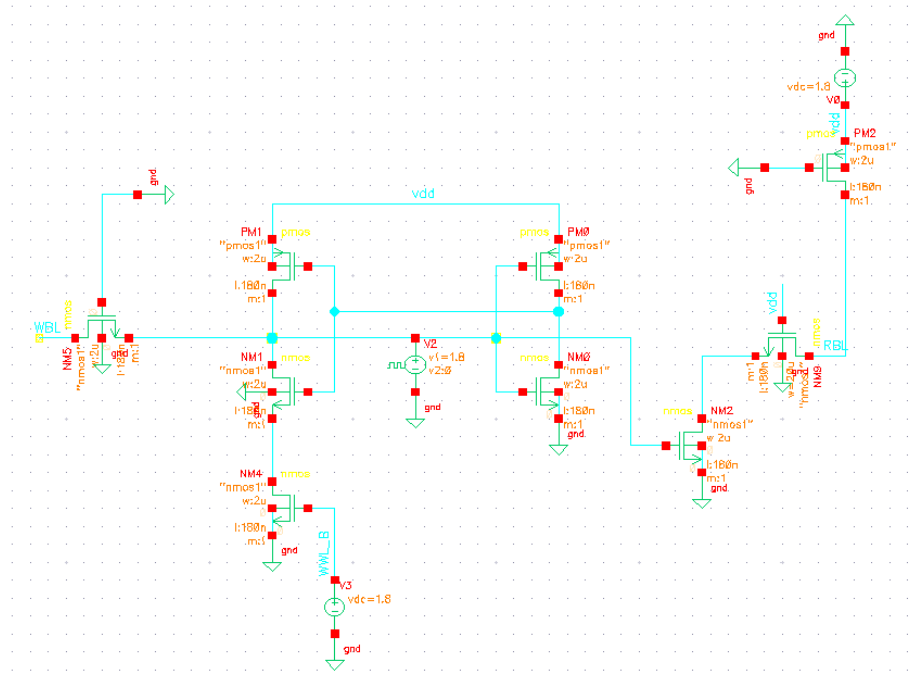


Figure 4: 8T SRAM cell Schematic in Cadence

The cell was implemented at the schematic level, ensuring correct read and write functionality under different input combinations. Special attention was given to sizing the transistors to balance performance and stability, especially in read-disturb scenarios.

4.2 Multi-Logic Sense Amplifier (MLSA) Design

The MLSA circuit is designed to sense small voltage differences on the read bit-line (RBL) and convert them into distinct logic outputs. The amplifier is capable of generating OR/NOR, AND/NAND, and XOR/XNOR outputs using a combination of latch-based sensing and inverter-based extensions.

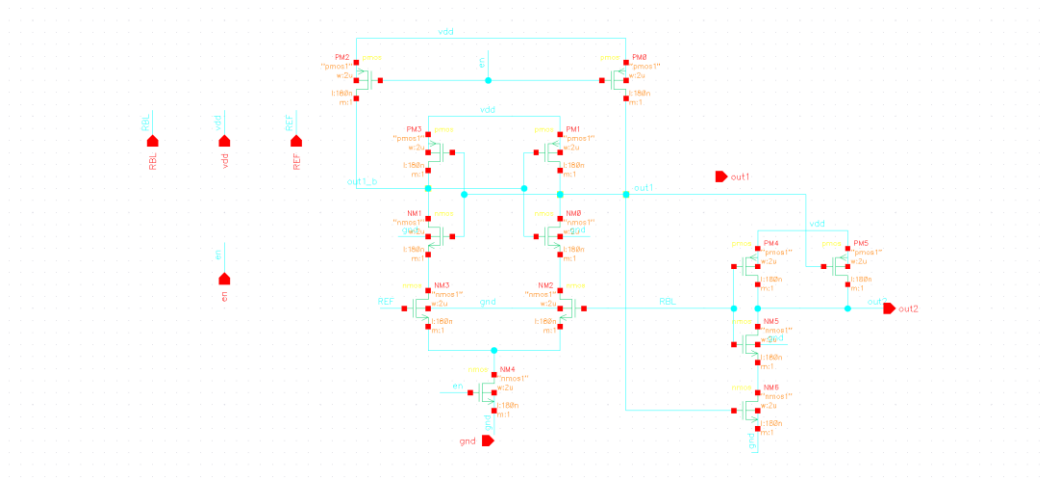


Figure 5: MLSA Schematic in Cadence

4.3 In-Memory computing Cell Design

For In-Memory computing cell 2 8T SRAM cells and MLSA is used to perform the computation. After that testbench is design to perform simulations and get transient wavefoorms.

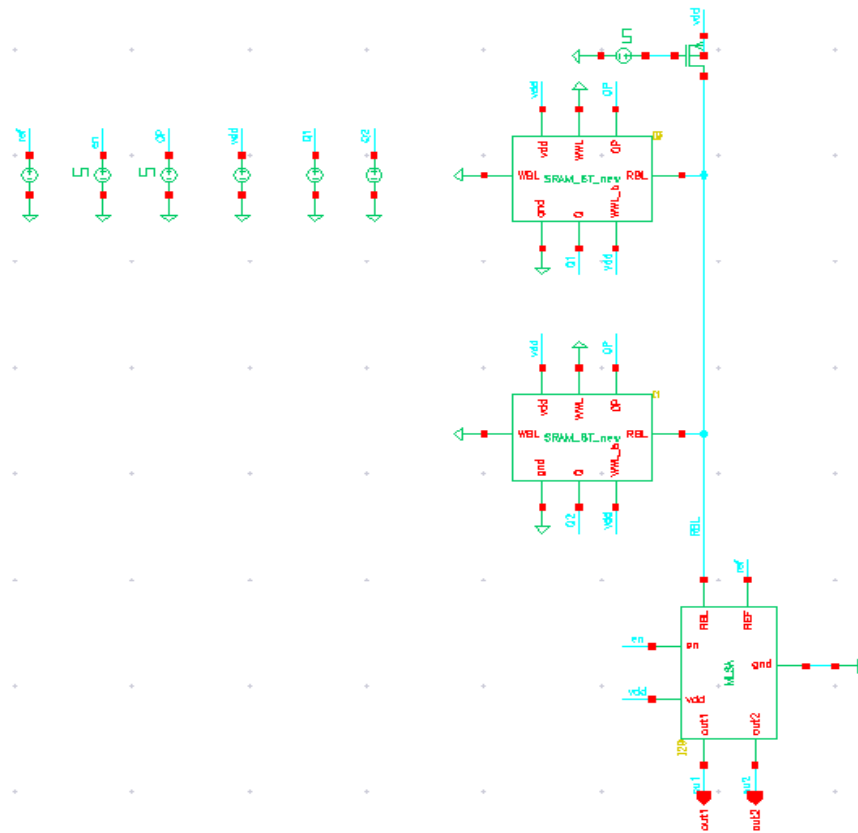


Figure 6: IMC Cell Schematic in Cadence

4.4 Layout Design

After successful schematic verification, the critical modules 8T SRAM cell, MLSA, and supporting logic were designed at the layout level using Cadence Virtuoso Layout Editor. The layout design followed Design Rule Checks (DRC) and Layout Versus Schematic (LVS) verification to ensure physical correctness and functional equivalence.

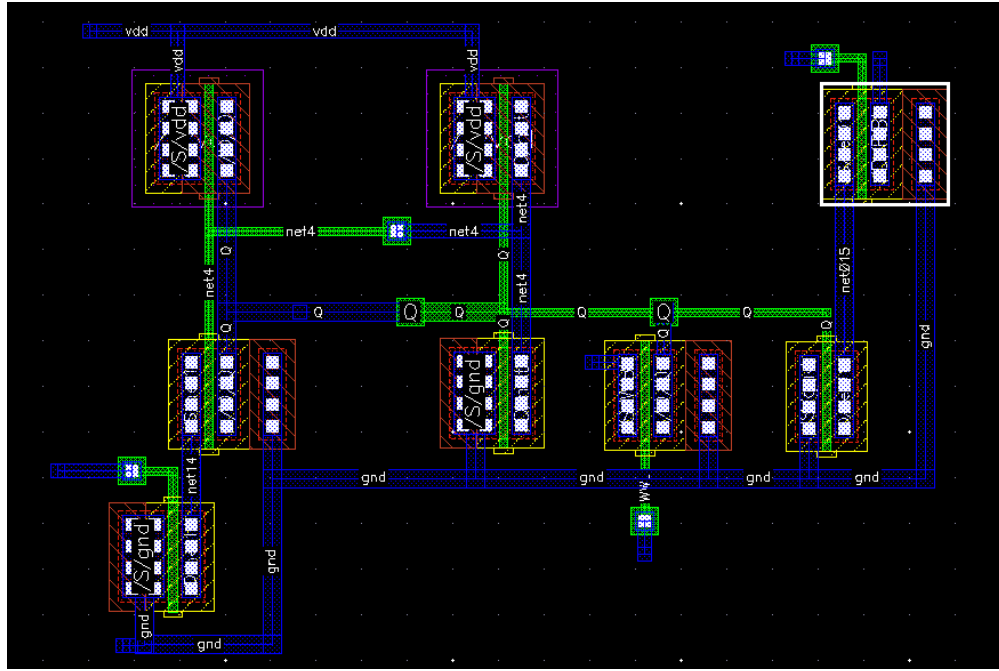


Figure 7: 8T SRAM Cell Layout

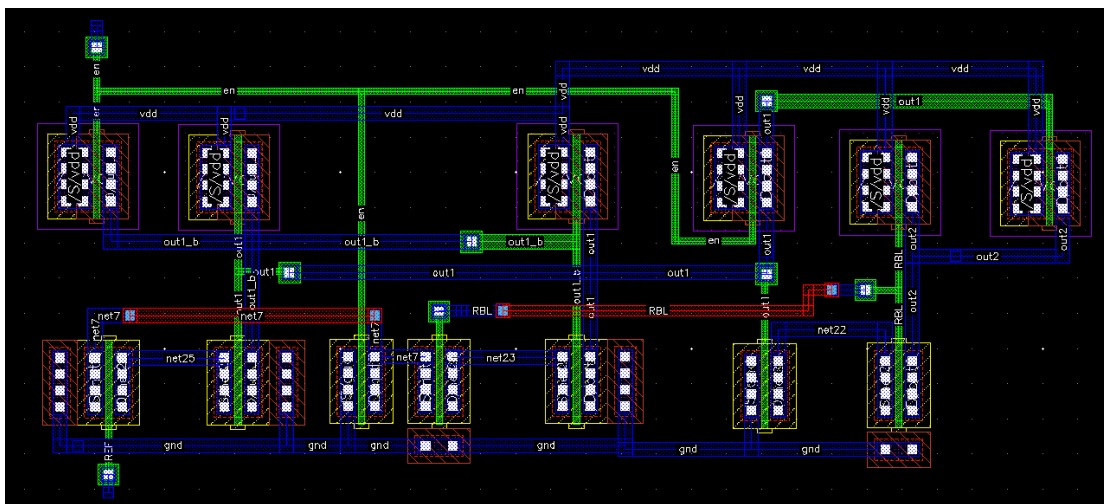


Figure 8: MLSA Layout

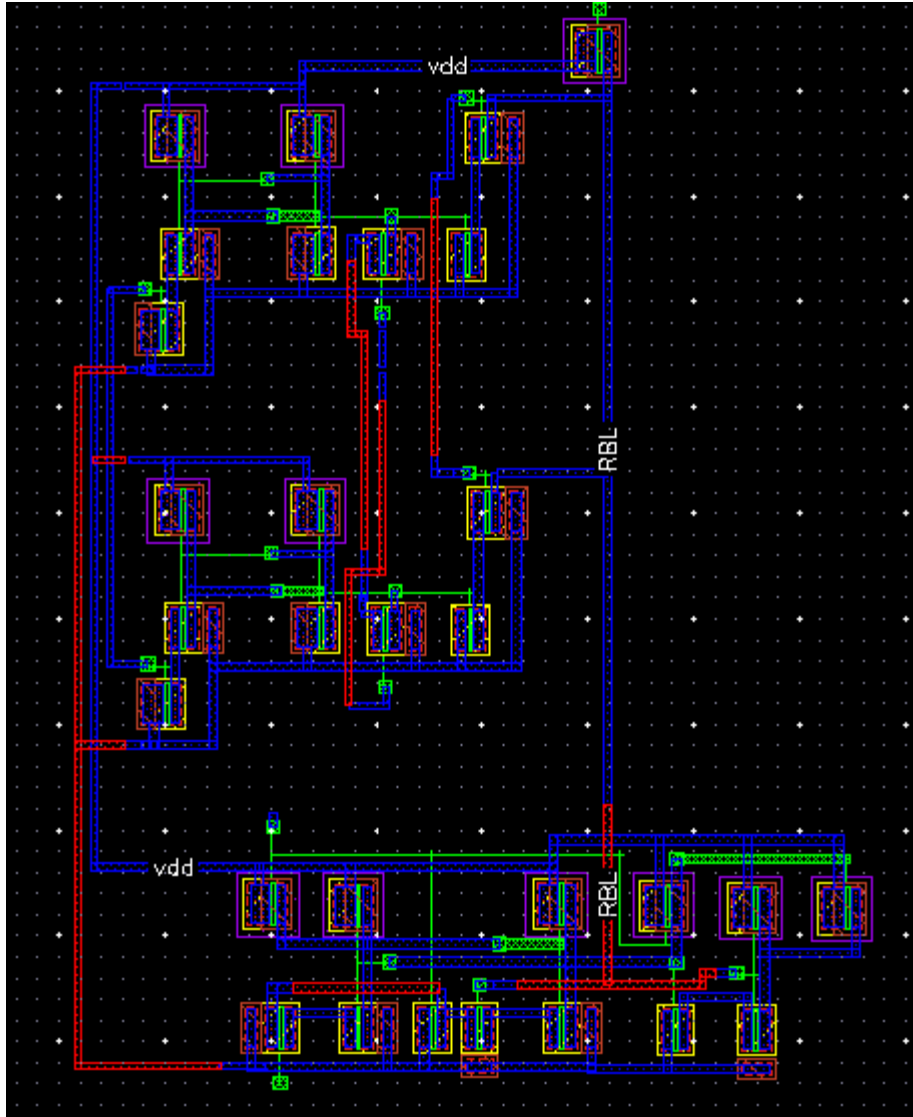


Figure 9: In-Memory computing cell layout

Chapter 5

RESULTS AND DISCUSSIO

5.1 8T SRAM Cell Read and Write Operations

(1) Read operation:

Before the operation, the RBL is pre-charged to VDD. During the read operation, the OP control signal of the target cell is set to 1, transistor M7 is turned on, and M8 is turned on or off, according to the stored data. And the RBL generates a discharge path at the read port of this cell, causing the RBL to reach a low-voltage state eventually.

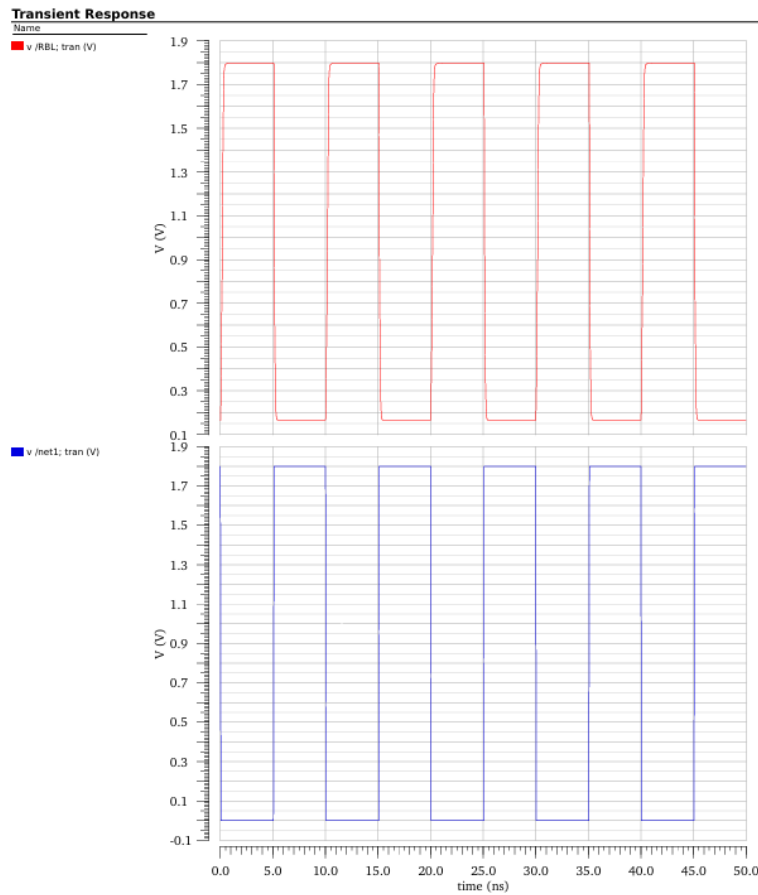


Figure 10: READ operation of 8T SRAM Cell

(2) Write Operation:

When performing a write operation, the write value is set to the WBL, and then the write word line (WWL) and $\sim$ WWL of the target cell are set to 1 and 0, respectively, so that the WBL value is written to the cell through transistor M6.

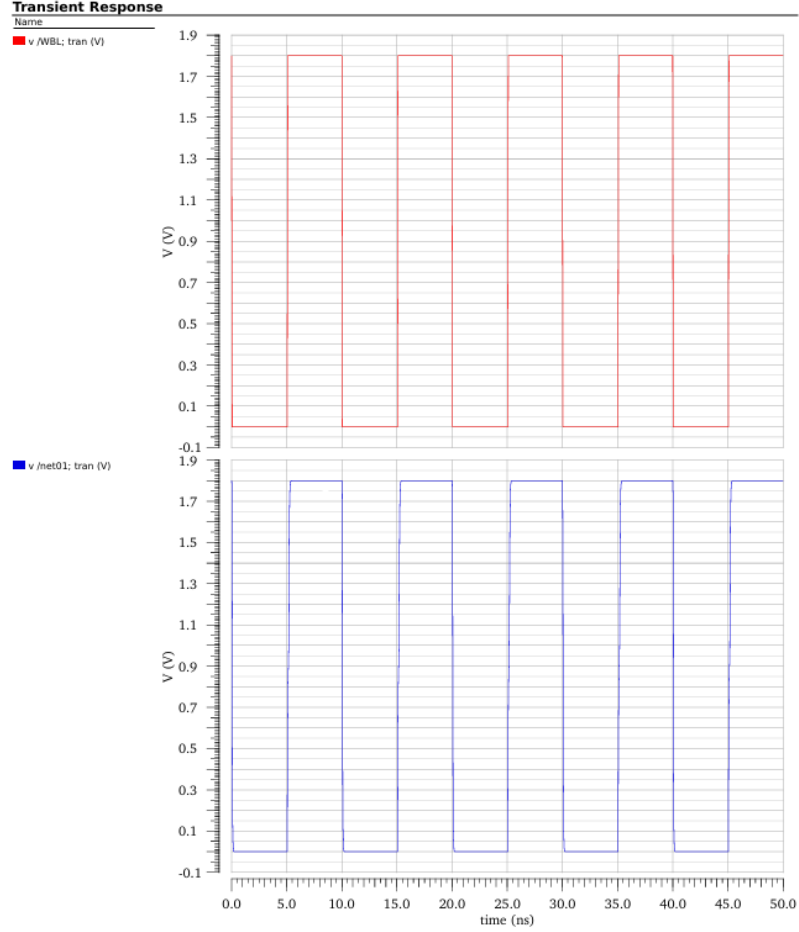


Figure 11: WRITE operation of 8T SRAM Cell

5.2 Bit-Line Discharge Rate

(1) Impact of Reference Voltage on Bit-Line Discharge

To assess how the sense amplifier performs under various reference voltages, simulations were conducted with differing V_{ref} values. The output discharge curves indicate that increasing the reference voltage accelerates the discharge on the read bit-line (RBL). Conversely, when V_{ref} drops below the threshold voltage of the NMOS transistor, the discharge process slows significantly. This degradation in discharge behavior negatively impacts the accuracy and speed of logic detection by the sense amplifier.

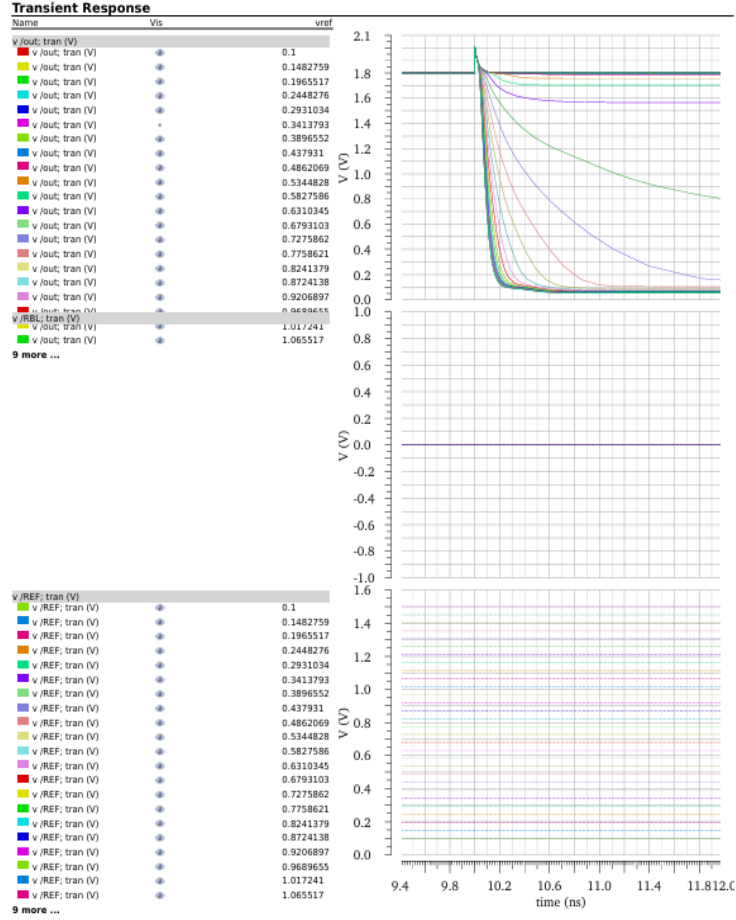


Figure 12: Bit-Line discharge rate with different Vref

(2) Effect of Stored Data on Bit-Line Discharge

Simulations also explored how different stored input patterns affect the discharge characteristics. A sense amplifier enable window of 12 ps was identified as optimal. Within this duration, the RBL voltage remains above $V_{DD}/2$ for intermediate input states (e.g., 01 and 10) and falls below $V_{DD}/2$ when both cells store '1' (i.e., 11). This separation in voltage levels enhances the reliability of logic recognition and improves the sensing yield. Proper timing control ensures clear distinction between logic states during in-memory computation.

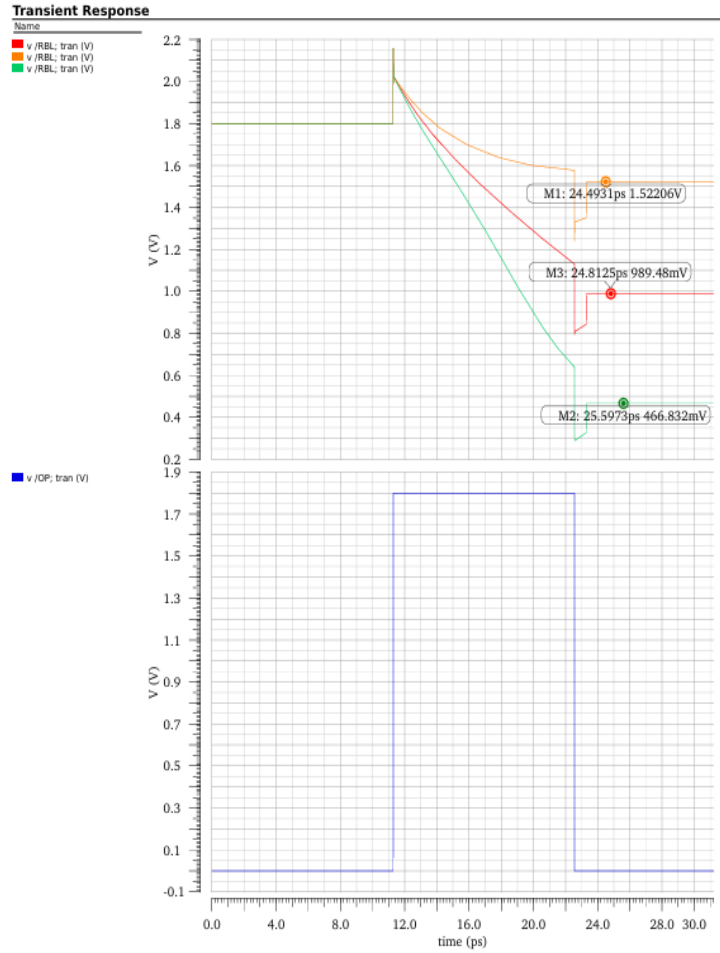


Figure 13: Bit-Line discharge rate with different data stored

5.3 Latch type Sense amplifier and MLSA Operations

Latch type sense amplifier performs only NOR and NAND operation and it requires two reference voltages Vref High and Vref Low. To perform XOR/XNOR operations additional circuits requires which increases the delay and power requirements.

Table 2: NOR/OR operation of conventional amplifier

A	B	RBL Voltage	Out(NOR)	~Out(OR)
0	0	$V_{rbl} > V_{ref\ High}$	1	0
0	1	$V_{rbl} < V_{ref\ High}$	0	1
1	0	$V_{rbl} < V_{ref\ High}$	0	1
1	1	$V_{rbl} < V_{ref\ High}$	0	1

Table 3: NAND/AND operation of conventional amplifier

A	B	RBL Voltage	Out(NAND)	~Out(AND)
0	0	$V_{rbl} > V_{ref} \text{ Low}$	1	0
0	1	$V_{rbl} > V_{ref} \text{ Low}$	1	0
1	0	$V_{rbl} > V_{ref} \text{ Low}$	1	0
1	1	$V_{rbl} < V_{ref} \text{ Low}$	0	1

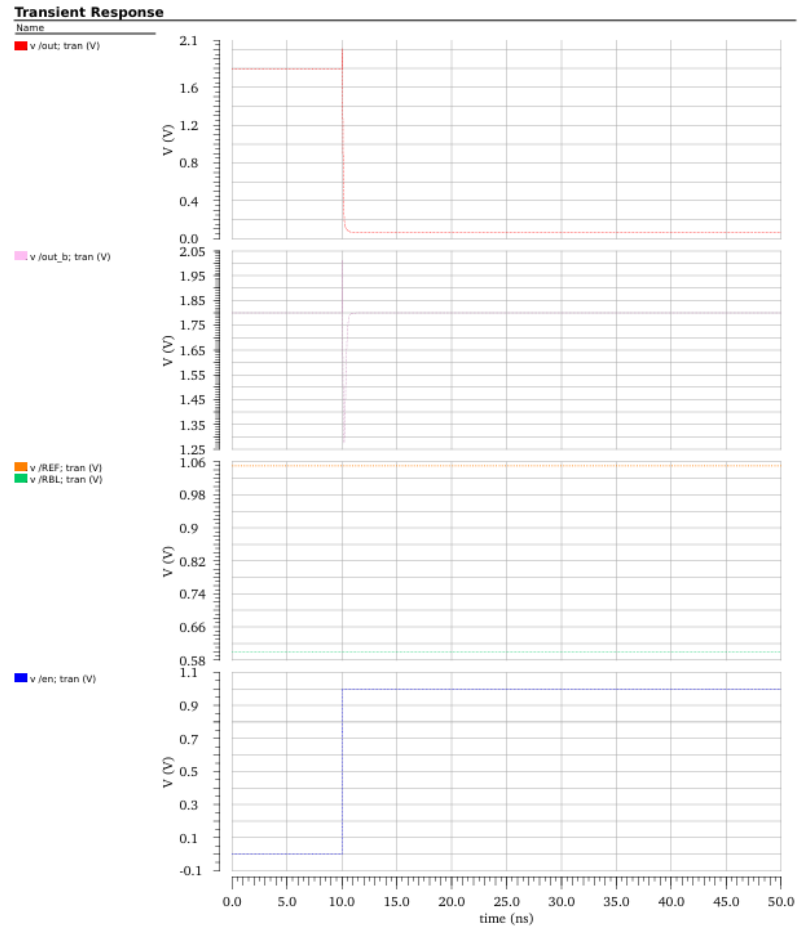


Figure 14: conventional sense amplifier NOR operation

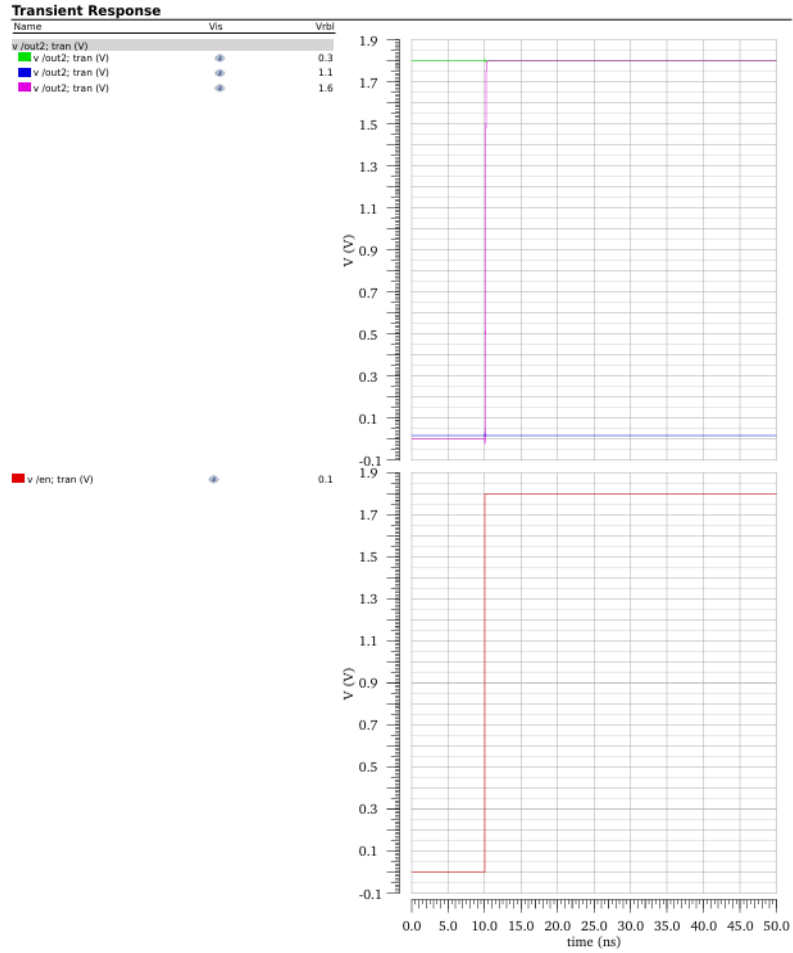


Figure 15: MLSA XNOR operation

5.4 XNOR and NOR operation of IMC Cell

XNOR and NOR operations are performed by IMC Cell, Bit line is precharged before performing the computing. XNOR operation takes less delay compare to conventional sense amplifier. In figure 15 XNOR operations is performed based on value stored in two 8T SRAM cell (00, 01/10 and 11). In figure 16 the NOR operation is performed using IMC cell. The OP is High for 22 ps for bit line discharge purpose.

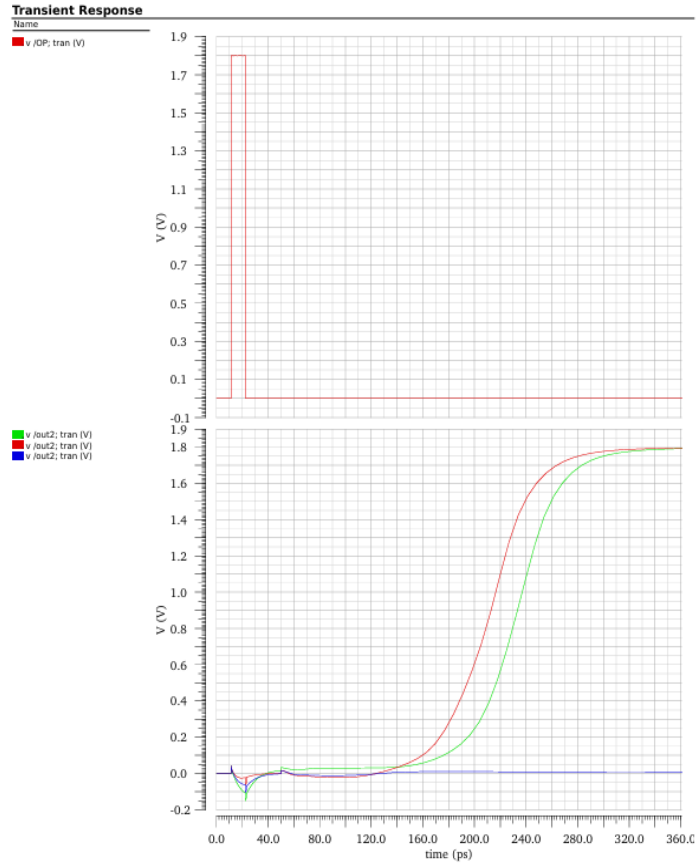


Figure 16: XNOR operation in IMC

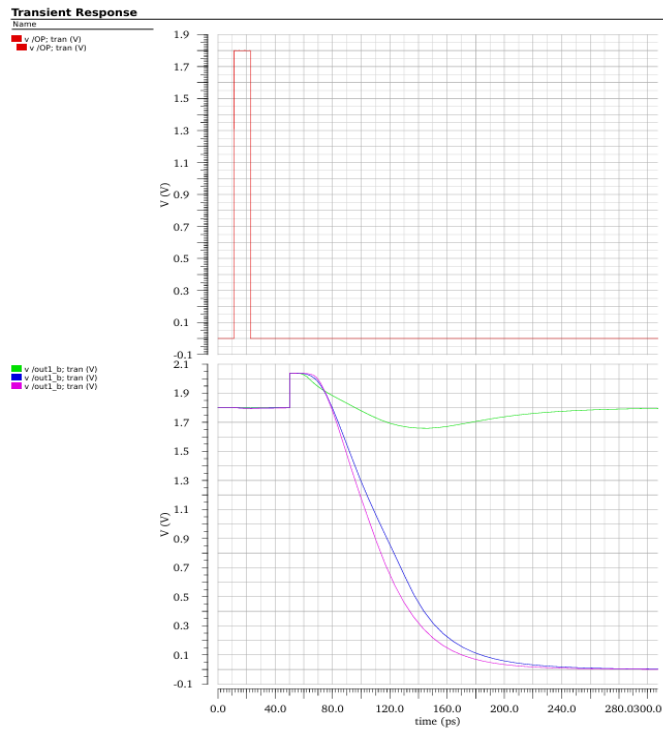


Figure 17: NOR operation in IMC

5.5 Monte Carlo analysis

To assess the robustness of the design under process variations, Monte Carlo simulations were conducted. Parameters such as threshold voltage and carrier mobility were varied randomly. The output confirmed consistent performance across hundreds of simulation iterations, validating the stability and reliability of the proposed IMC cell architecture.

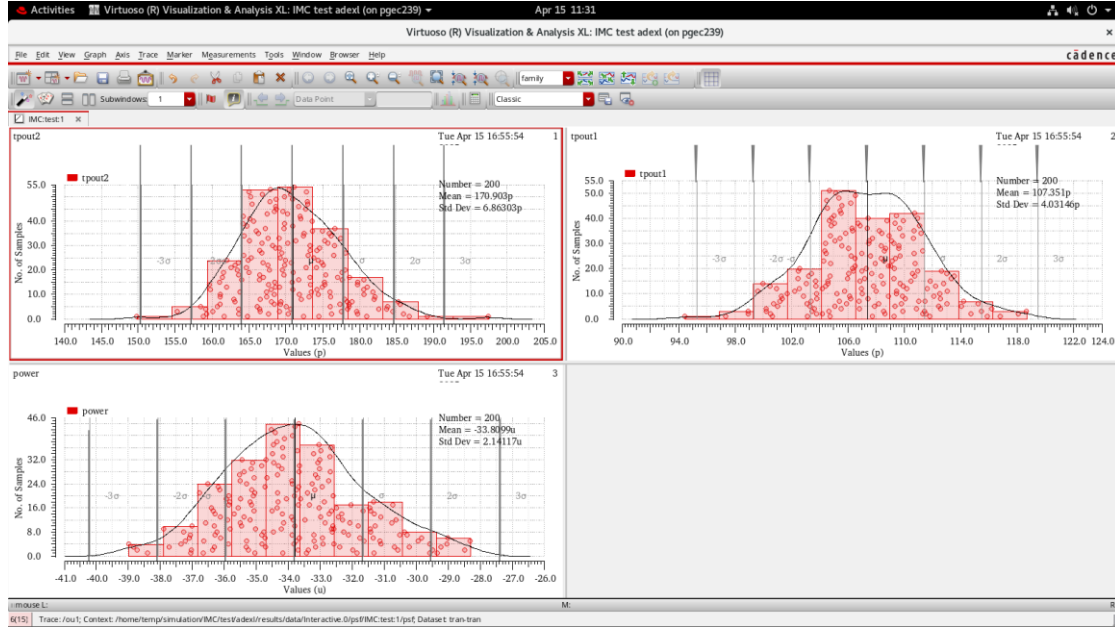


Figure 18: Monte Carlo simulation of Power, Delay of OR and XNOR operation for case (00)

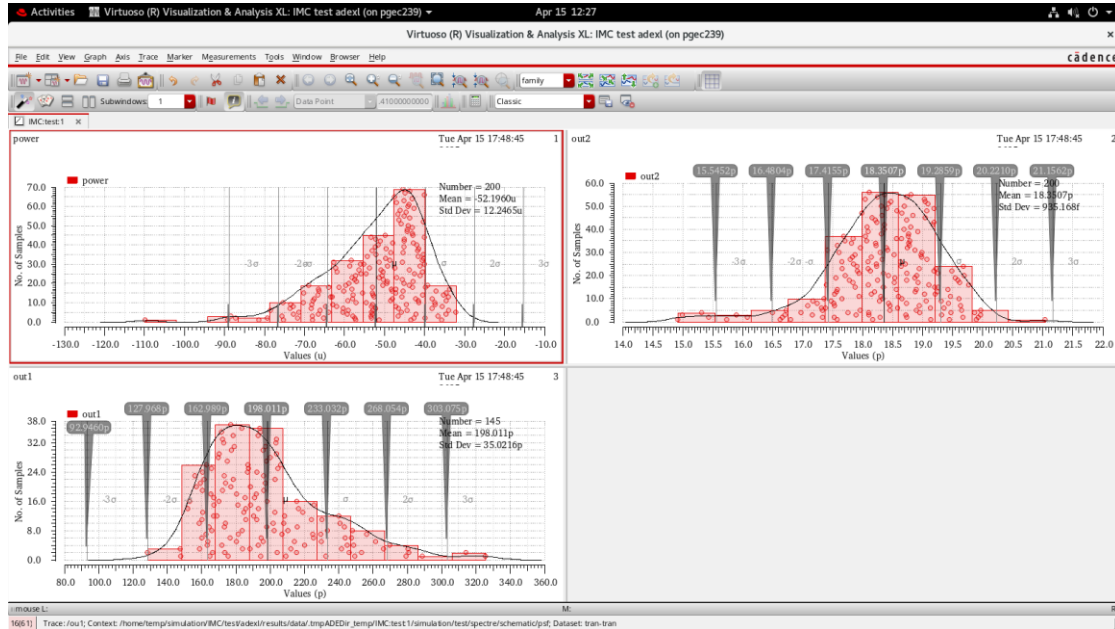


Figure 19: Monte Carlo simulation of Power, Delay of OR and XNOR operation for case (01/10)

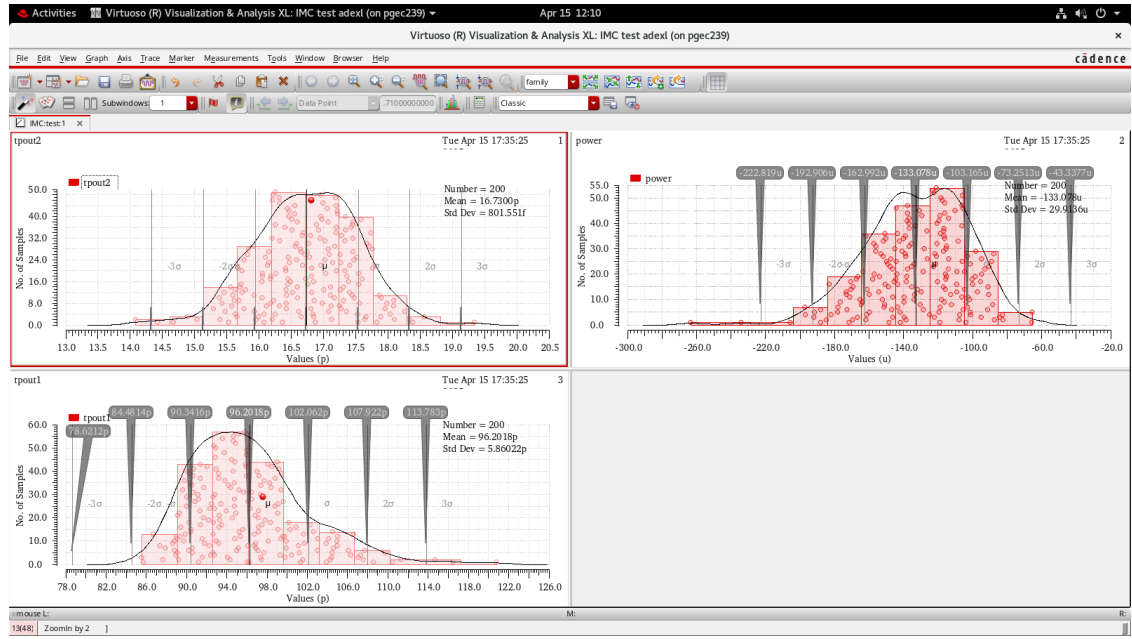


Figure 20: Monte Carlo simulation of Power, Delay of OR and XNOR operation for case (11)

5.6 Power, Delay Analysis

Table 4: Power, Delay and PDP for XNOR operation

Cases	Power(μ W)	Delay(pS)	PDP(f)
00	33.80	170.90	5.77
01/10	52.196	18.35	0.95
11	133.1	16.75	2.29

Table 5: Power, Delay and PDP for OR operation

Cases	Power(μ W)	Delay(pS)	PDP(f)
00	33.80	107.65	3.64
01/10	52.196	198.011	10.33
11	133.1	96.2	12.80

Chapter 6

CONCLUSION

In this project, a novel in-memory computing architecture based on an 8-transistor (8T) SRAM cell integrated with a Multi-Logic Sense Amplifier (MLSA) was designed and evaluated using 180nm CMOS technology. The proposed system was capable of performing essential logic operations such as OR, NOR, XOR, and XNOR within the memory itself, effectively reducing the need for data transfer between the memory and processing units. This paradigm shift addresses the key bottleneck in conventional von Neumann architectures namely, the latency and energy overhead associated with memory access.

The use of an 8T SRAM structure proved advantageous due to its separate read and write paths, which enhanced read stability and supported concurrent wordline activation. This facilitated voltage-based computation using bitline discharge characteristics without compromising the integrity of stored data. Furthermore, the MLSA was carefully designed to interpret voltage levels across the read bit-line using dynamically tunable reference voltages. This allowed the amplifier to support multiple logic functions within a single sense unit, significantly reducing hardware overhead and improving energy efficiency.

Simulation results obtained from Cadence Virtuoso demonstrated the correctness of logic operations under different input combinations. Performance analysis including transient response, delay, and power dissipation—confirmed that the proposed architecture offers notable improvements in terms of speed and energy consumption, especially when compared to conventional sense amplifiers. Monte Carlo analysis further validated the robustness of the design under process variations.

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